

MEASURING WEIGHT

Weight is a measure of how heavy an object is. In the customary system, weight is measured in the three basic units shown in the table.

Unit	Sample Use	Comparing Units
ounce (oz)		$1 \text{ oz} = \frac{1}{16} \text{ lb}$
pound (lb)		$1 \text{ lb} = 16 \text{ oz}$
ton (t)		$1 \text{ t} = 2,000 \text{ lb}$

Converting: Smaller ↔ Larger

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CONVERTING TO LARGER UNITS

Example Ibrahim bought a bag of candy weighing 45 oz. What is that weight in pounds and ounces?

Step 1

Set up the problem.

$$45 \text{ oz} = \underline{\hspace{1cm}} \text{ lb } \underline{\hspace{1cm}} \text{ oz}$$

The bag weighs **2 lb 13 oz**.

Step 2

Divide by the number of smaller units. (16 oz = 1 lb)

$$45 \text{ oz} \div 16 = 2 \text{ R}13$$

Step 3

Write the whole number as the larger unit and the remainder as the smaller unit.

$$45 \text{ oz} = 2 \text{ lb } 13 \text{ oz}$$

CONVERTING TO SMALLER UNITS

Example Heidi needed 60 oz of beef to serve her guests at a dinner party. She bought $3\frac{3}{4}$ lb of meat. Was that enough?

Step 1

Set up the problem.

$$3\frac{3}{4} \text{ lb} = \underline{\hspace{1cm}} \text{ oz}$$

Yes, Heidi has just enough meat (60 oz).

Step 2

To change to a smaller unit, multiply by the number of smaller units.

$$3\frac{3}{4} \times 16 = \frac{15}{1} \times \frac{16}{1} = 60 \text{ oz}$$

Change each amount into larger or smaller units as indicated.

1. $32 \text{ oz} = \underline{\hspace{1cm}} \text{ lb}$

$$92 \text{ oz} = \underline{\hspace{1cm}} \text{ lb } \underline{\hspace{1cm}} \text{ oz}$$

$$40 \text{ oz} = \underline{\hspace{1cm}} \text{ lb } \underline{\hspace{1cm}} \text{ oz}$$

$$4,500 \text{ lb} = \underline{\hspace{1cm}} \text{ t } \underline{\hspace{1cm}} \text{ lb}$$

2. $1\frac{1}{2} \text{ lb} = \underline{\hspace{1cm}} \text{ oz}$

$$2\frac{1}{4} \text{ lb} = \underline{\hspace{1cm}} \text{ oz}$$

$$3 \text{ t} = \underline{\hspace{1cm}} \text{ lb}$$

$$4\frac{3}{4} \text{ t} = \underline{\hspace{1cm}} \text{ lb}$$

Adding & Subtracting Weight

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ADDING OR SUBTRACTING UNITS OF WEIGHT

Example Cobi roasted 2 chickens. The first weighed 4 lb 10 oz. The second weighed 3 lb 14 oz. What was the chickens' combined weight?

Step 1

Set up the problem.

$$\begin{array}{r} 4 \text{ lb } 10 \text{ oz} \\ + 3 \text{ lb } 14 \text{ oz} \\ \hline \end{array}$$

Step 2

Estimate first by adding the larger units.

$$4 \text{ lb} + 3 \text{ lb} = 7 \text{ lb}$$

The answer should be slightly more than 7 lb.

Step 3

Add or subtract the units separately.

$$\begin{array}{r} 4 \text{ lb } 10 \text{ oz} \\ + 3 \text{ lb } 14 \text{ oz} \\ \hline 7 \text{ lb } 24 \text{ oz} \end{array}$$

Step 4

Regroup or simplify the answer, if necessary.

$$\begin{array}{r} 7 \text{ lb } 24 \text{ oz} = \\ 7 \text{ lb} + 1 \text{ lb } 8 \text{ oz} = \\ 8 \text{ lb } 8 \text{ oz} \end{array}$$

8 lb 8 oz is slightly more than 7 lb. The combined weight is **8 lb 8 oz**.

Add or subtract the following weights.

3.
$$\begin{array}{r} 2 \text{ lb } 3 \text{ oz} \\ + 1 \text{ lb } 8 \text{ oz} \\ \hline \end{array}$$

$$\begin{array}{r} 6 \text{ lb } 14 \text{ oz} \\ + 4 \text{ lb } 11 \text{ oz} \\ \hline \end{array}$$

$$\begin{array}{r} 2 \text{ t } 152 \text{ lb} \\ + 1 \text{ t } 576 \text{ lb} \\ \hline \end{array}$$

$$\begin{array}{r} 3 \text{ t } 1,500 \text{ lb} \\ + 5 \text{ t } 1,250 \text{ lb} \\ \hline \end{array}$$

4.
$$\begin{array}{r} 5 \text{ lb } 12 \text{ oz} \\ - 3 \text{ lb } 8 \text{ oz} \\ \hline \end{array}$$

$$\begin{array}{r} 7 \text{ lb } 3 \text{ oz} \\ - 5 \text{ lb } 12 \text{ oz} \\ \hline \end{array}$$

$$\begin{array}{r} 12 \text{ lb } 14 \text{ oz} \\ - 6 \text{ lb } 15 \text{ oz} \\ \hline \end{array}$$

$$\begin{array}{r} 4 \text{ t } 800 \text{ lb} \\ - 1 \text{ t } 1,200 \text{ lb} \\ \hline \end{array}$$

Applications

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Solve the following word problems.

5. Trevor used about half of a 5-lb bag of potatoes (2 lb 6 oz). How much did the remaining potatoes weigh?
6. Tyus can carry $1\frac{1}{2}$ t of cargo in his truck. He loads the truck with 1,756 lb of bricks. How many more pounds of bricks can Tyus put in his truck?
7. McGarity's Bar and Grill had 35 lb 3 oz of steak left over from last night. They received a delivery of 14 lb 15 oz today. How much steak do they have altogether?
8. Keiko is using a forklift to load shipping crates on a freight elevator. Each crate weighs 220 lb. The elevator can hold 1 t safely. Can she safely load 9 crates?
9. **Discuss** Which customary weight units would you use to measure a truck? A man? A box of cereal?